# Annex D

## FORMANT3.M Program Code (*Matlab 4.0*)

```
function formant3(SpecFile,PlotFlag,NRFlag,NoiseFile)

% YOUNG FORMANT EXTRACTOR
%
% PROGRAM OVERVIEW:
%
% USAGE: formant3(SpecFile,PlotFlag,NRFlag,NoiseFile)
%
% Plot Flag is used to control the level of screen "visualization"
% PlotFlag==2 will plot 9 figures
% PlotFlag==1 will plot 1 figure (the most important) of filtered speech formants
% PlotFlag==0 plots nothing (fastest execution)
%
% NRFlag==1 performs noise reduction of the spectrogram using the wav file "NoiseFile"
%       and is highly recommended for steady state background noise (ie. hard disk RPMs
etc)

% TURN ON WHITE BACKGROUND FOR SCREEN SHOTS
% ensure all figures are closed first
close; close; close; close; close; close; close; close; close; close; close;
whitebg;

%##########################################################################
%       PART 1 - LOAD SPECTROGRAM
%##########################################################################
```

```
load(SpecFile);
[SpecRows,SpecCols]=size(Spec);

%~~~~~~~~~~~~~~ GLOBAL CONSTANTS ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

NFFT=512;                   % Number of samples in FFT window frame
NOverLap=0.5*NFFT;
FreqRes=Fs/NFFT;    % Bandwidth resolution of FFT frequency bands (specgram rows)
MinFreq=300;                % Ignore frequencies below this level
MaxFreq=3300;               % Ignore frequencies above this level
                            % Eg. Telephone bandwidth 300-3300 Hz
% The specgram row representing the miminum frequency
LowBand=ceil(MinFreq/FreqRes);
% The specgram row representing the maximum frequency
HighBand=ceil(MaxFreq/FreqRes);
% The number of formants to be extracted from the signal
MaxFormants=3;
% The Half Formant Bandwidth in frequency (Hz)
HFBFreq=150;
% The Half Formant Bandwidth in specgram rows
HFBRows=ceil(HFBFreq/FreqRes);

%~~~~~~~~~~~~~~ GLOBAL CONSTANTS ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

% PLOT THE SPECTROGRAM OF JUST LOADED
if PlotFlag > 1
        figure(1)
        clf reset
        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
```

```
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec); axis('xy')
        title(['Spectrogram of ',WaveFile]) ;
        xlabel('Time (sec)');
        ylabel('Frequency (Hz)');
        colormap(1-gray);
      pause(3);
end;


if NRFlag > 0
  % PRODUCE A NOISE REDUCTION MASK
  [NOISE]=loadwav(NoiseFile); % background noise recorded from live session
  MaxNoise=max(abs(specgram(NOISE*Boost_Ratio,NFFT))');
  % max of transpose specgram matrix gives freq row maximums
  MaxNoise=MaxNoise'; % convert to a column vector
  MaxNoise=MaxNoise(1:SpecRows,1); % crop to same number of rows as spectrogram
  if PlotFlag > 1
        % PLOT MAXNOISE
        figure(2)
        clf reset;
        TimeIndex=[0 1];
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,MaxNoise); axis('xy')
        title(['Max Spectral Amplitudes of ',NoiseFile]);
        ylabel('Frequency (Hz)');
        colormap(1-gray);
        pause(3);
  end; % if PlotFlag

  for TimeIndex=1:SpecCols
```

```
        Spec(:,TimeIndex)=Spec(:,TimeIndex)-MaxNoise;
  end;

  % ELIMINATE NEGATIVE NUMBERS FROM SPECTROGRAM
  Spec=(Spec+abs(Spec))/2;

  if PlotFlag > 1
        figure(3)
        clf reset;
        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec); axis('xy')
        title(['Noise Masked Spectrogram of ',WaveFile]) ;
        xlabel('Time (sec)');
        ylabel('Frequency (Hz)');
        colormap(1-gray);
        pause(3);
  end;
end; % if NoiseReduction

%##########################################################################
%
% PART 2 - PROCESS SPECTROGRAM FOR VOICED PHONEMES IN UTTERANCE
%
%##########################################################################

% DETERMINE SHORT TERM ENERGY OF SPEC COLUMNS WITHIN THE GIVEN
% FREQUENCY RANGE
STEnergy=(sum(Spec(LowBand:HighBand,:)));
```

```
% PLOT SHORT TERM ENERGY CURVE ON SPECTROGRAM (SCALED TO MAX
% FREQUENCY)
if PlotFlag > 1
        figure(4)
        clf reset;

        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec(1:HighBand,:)); axis('xy')
        title(['Relative STE Curve of ',WaveFile]) ;
        xlabel('Time (sec)');
        ylabel('Frequency (Hz)');
        colormap(1-gray);

        % Plot STE curve overlay
        hold on;
        plot(TimeIndex,max(FreqIndex)*STEnergy/max(STEnergy),'k')
        hold off

        % mark the minimum and maximum frequency band stops
        hold on;
        plot(TimeIndex,MinFreq*ones(1,SpecCols),'k');
        plot(TimeIndex,MaxFreq*ones(1,SpecCols),'k');
        hold off;
        pause(3);
end;

% REMOVE PORTIONS OF LOW ENERGY (SILENCE AND FRICATIVES)
% using 2nd frequency row as a reference
Index=find(HighBand*STEnergy/max(STEnergy) > 2);
```

```
% keep only the high energy portions
Spec=Spec(:,Index);
% determine new size of spectrogram
[SpecRows,SpecCols]=size(Spec);
% keep only the high energy portions
STEnergy=STEnergy(Index);
LengthSTE=length(STEnergy);

% PLOT SPECTROGRAM WITH LOW ENERGY SECTIONS REMOVED
if PlotFlag > 1
        figure(5)
        clf reset;
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec(1:HighBand,:)); axis('xy')
        title(['Silence Reduced Spectrogram of ',WaveFile]) ;
        xlabel('Time (sec)');
        ylabel('Frequency (Hz)');
        colormap(1-gray);

        % Plot STE curve overlay
        hold on;
        plot(TimeIndex,max(FreqIndex)*STEnergy/max(STEnergy),'k')
        hold off

        % mark the minimum and maximum frequency band stops
        hold on;
        plot(TimeIndex,MinFreq*ones(1,SpecCols),'k');
        plot(TimeIndex,MaxFreq*ones(1,SpecCols),'k');
        hold off;
```

```
        pause(3);
end;


% APPLY OCTAVE GAIN TO EMPHASIZE HIGHER FREQUENCY BANDS
for row=1:SpecRows
        Gain(row,1)=10^(log2(row)/5); %
end;
for TimeIndex=1:SpecCols
        Spec(:,TimeIndex)=Spec(:,TimeIndex).*Gain;
end;

if PlotFlag > 1
        % PLOT SPECTROGRAM TO SCREEN
        figure(6);
        clf reset;

        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec(1:HighBand,:)); axis('xy')
        title(['Octave Enhanced Spectrogram of ',WaveFile]) ;
        xlabel('Time (sec)');
        ylabel('Frequency (Hz)');
        colormap(1-gray);

        % mark the minimum and maximum frequency band stops
        hold on;
        plot(TimeIndex,MinFreq*ones(1,SpecCols),'k');
        plot(TimeIndex,MaxFreq*ones(1,SpecCols),'k');
        hold off;
```

```
        pause(3);
end;
%##########################################################################
%
%       PART 3 - EXTRACT FORMANTS
%
%##########################################################################

% PERFORM INITIALIZATIONS BASED ON SIZE OF SPECTROGRAM
Peak=zeros(SpecRows,SpecCols);
Formant=zeros(SpecRows,SpecCols);


% DETERMINE SPECTRAL PEAKS FOR EACH TIME SLICE

for TimeIndex=1:SpecCols

  PeakNumber=0;

  % SAVE FREQUENCY BIN NUMBER OF LOCAL PEAKS
  % (max values within Formant Bandwidth)
  for FreqBand=LowBand:HighBand;
    LowFB=FreqBand-HFBRows;
    if LowFB < 1
        LowFB=1;
    end;

    HighFB=FreqBand+HFBRows;
    if HighFB > SpecRows
        HighFB=SpecCols;
```

```
    end;

    FBMax=max(Spec(LowFB:HighFB,TimeIndex)); % the max value within the window

    if Spec(FreqBand,TimeIndex) == FBMax & FBMax ~= 0
        PeakNumber=PeakNumber+1;
        % record the frequency band of the local peak
        PeakFrequency(PeakNumber,TimeIndex)=FreqBand;
        % record its amplitude
        PeakAmplitude(PeakNumber,TimeIndex)=Spec(FreqBand,TimeIndex);
        % keep track of the total number of peaks in a given time slice
        NumberOfPeaks(TimeIndex)=PeakNumber;
    end;
  end; % for FreqBand

  % SINCE ARRAY WAS NOT INITIALIZED, TIME INDEX LENGTH IS VARIABLE
  % SINCE PEAKS MAY NOT BE PRESENT AT THE  END OF THE
 % SPECTROGRAM. IN THIS CASE, INSERT DUMMY VALUES AS PLACE
  % HOLDERS TO PROVIDE UNIFORM ARRAY LENGTHS
  if PeakNumber==0
        PeakFrequency(1,TimeIndex)=1;
        PeakAmplitude(1,TimeIndex)=1;
        NumberOfPeaks(TimeIndex)=0;
  end;
end; % for TimeIndex


% DEFINE FORMANTS AS PEAKS WITH GREATEST AMPLITUDE / NUMBERED
% FROM LOW TO HIGH FREQUENCY
for TimeIndex=1:SpecCols
        % SORT AMPLITUDES (ascending order)
```

```
        [SortedAmplitudes,PeakNumbers]=sort(PeakAmplitude(:,TimeIndex));
        % change the index to descending order
        PeakNumbers=flipud(PeakNumbers);

        % SELECT THE ONES WITH GREATEST AMPLITUDES
        if NumberOfPeaks(TimeIndex) < MaxFormants
                % pick only non-zero elements
                MaxPeaks=PeakNumbers(1:NumberOfPeaks(TimeIndex));
        else
                % pick the biggest ones in the index
                MaxPeaks=PeakNumbers(1:MaxFormants);
        end; % if

% LABEL THE FORMANTS IN ORDER OF PEAK NUMBER (LOWEST TO HIGEST
% FREQUENCY BAND)
        SortedMaxPeaks=sort(MaxPeaks);
        for FormantNumber=1:length(MaxPeaks)
        Formant(FormantNumber,TimeIndex)=
                PeakFrequency(SortedMaxPeaks(FormantNumber),TimeIndex);
        end;

end; % for TimeIndex

if PlotFlag > 1
        % PLOT SPECTROGRAM TO SCREEN
        figure(7);
        clf reset;

        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
```

```
        imagesc(TimeIndex,FreqIndex,Spec(1:HighBand,:)); axis('xy')
        title(['Local Spectral Maxima of ',WaveFile]) ;
        xlabel('(Symbol Key)    . = PEAK    X = F1    O = F2    * = F3');
        ylabel('Frequency (Hz)');
        colormap(1-gray);

        % mark the minimum and maximum frequency band stops
        hold on;
        plot(TimeIndex,MinFreq*ones(1,SpecCols),'k');
        plot(TimeIndex,MaxFreq*ones(1,SpecCols),'k');

        % PLOT PEAKS AND ROUGH DRAFT OF FORMANTS ON SPECTROGAM
        plot(TimeIndex,PeakFrequency'*FreqRes-(FreqRes/2),'.k')
        plot(TimeIndex,Formant(1,:)'*FreqRes-(FreqRes/2),'xk')
        plot(TimeIndex,Formant(2,:)'*FreqRes-(FreqRes/2),'ok')
        plot(TimeIndex,Formant(3,:)'*FreqRes-(FreqRes/2),'*k')

        hold off;
        pause(3);
end;


%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%   FILTER OUT SECTIONS WITH HIGH FIRST FORMANT
%   OR NO THIRD FORMANT
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% FIND ALL ZERO VALUED THIRD FORMANTS AND SET ALL THE FORMANTS
% FOR THAT TIME SLICE TO ZERO
ZeroIndex=find(Formant(3,:) == 0);
for TimeIndex=1:length(ZeroIndex)
```

```
   for FormantNumber=1:MaxFormants
        Formant(FormantNumber,ZeroIndex(TimeIndex))=0;
   end;
end;

% FIND ALL FIRST FORMANTS GREATER THAN 1200 HZ AND ZERO OUT THE
TIMESLICES
ZeroIndex=find(Formant(1,:) > ceil(1200/FreqRes));
for TimeIndex=1:length(ZeroIndex)
   for FormantNumber=1:MaxFormants
        Formant(FormantNumber,ZeroIndex(TimeIndex))=0;
   end;
end;


%%%%%%%%%%%%%%%%% PLOT FILTERED FORMANTS
if PlotFlag > 1
        figure(8);
        clf reset;
        [SpecRows,SpecCols]=size(Spec);
        TimeIndex=(1:SpecCols)*NOverLap/Fs;
        FreqIndex=(1:SpecRows)*Fs/NFFT;
        imagesc(TimeIndex,FreqIndex,Spec(1:HighBand,:)); axis('xy')
        title(['Probable Formants of ',WaveFile]) ;
        xlabel('(Symbol Key)    X = F1    O = F2    * = F3');
        ylabel('Frequency (Hz)');
        colormap(1-gray);
        % mark the minimum and maximum frequency band stops
        hold on;
        plot(TimeIndex,MinFreq*ones(1,SpecCols),'k');
        plot(TimeIndex,MaxFreq*ones(1,SpecCols),'k');
```

```
        % plot filtered formants
        plot(TimeIndex,Formant(1,:)'*FreqRes-(FreqRes/2),'xk')
        plot(TimeIndex,Formant(2,:)'*FreqRes-(FreqRes/2),'ok')
        plot(TimeIndex,Formant(3,:)'*FreqRes-(FreqRes/2),'*k')

        hold off;
        pause(3);
end;

%%%% KEEP ONLY THOSE FORMANT TIMESLICES WITH NON-ZERO VALUES
Formant=Formant(1:MaxFormants,find(Formant(3,:) > 0));
TimeIndex=(1:length(Formant))*NOverLap/Fs;


%%%%%%%%%%%%%% CONVERT FORMANT VALUES TO FREQUENCY
Formant=Formant*FreqRes-(FreqRes/2);

if PlotFlag > 0
        % PLOT FORMANTS ON FREQUENCY SCALE FOR INSPECTION
        figure(9)
        clf reset;
        plot(TimeIndex,Formant(1,:)','k');
        title(['Formant Vectors from ',WaveFile]);
        xlabel('(Symbol Key)    X = F1    O = F2    * = F3');
        ylabel('Frequency (Hz)');
        hold on;
        plot(TimeIndex,Formant(2,:)','k');
        plot(TimeIndex,Formant(3,:)','k');
        plot(TimeIndex,Formant(1,:)','kx');
        plot(TimeIndex,Formant(2,:)','ko');
```

```
        plot(TimeIndex,Formant(3,:)','k*');
        hold off;
        grid on;

        pause(3);
end;

%##########################################################################
%
%       PART 5 - SAVE RESULTS TO FILE
%
%##########################################################################

% SAVE FEATURE VECTOR IN FORMANT "F" FILE

save tempfile Formant WaveFile
clear
load tempfile
save(['f',WaveFile(2:length(WaveFile)-4),'.mat'])
fprintf('Processing Completed !! \nResults stored in %s \n',
['f',WaveFile(2:length(WaveFile)-4),'.mat']);
```